

STM32配置摄像头

注意：esp32摄像头需要烧录出厂固件，esp32摄像头到手之后没有刷过固件的则不需要，出厂默认的出厂固件，使用iic通信前，可以先使用串口给esp32摄像头配置上网络，iic用作数据读取

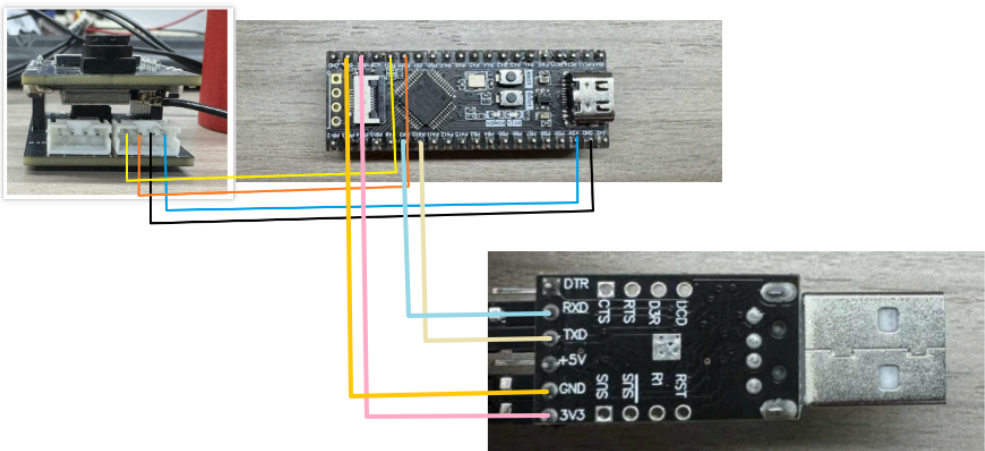
1.实验准备

- mspm0g350系列型号单片机
- wifi摄像头

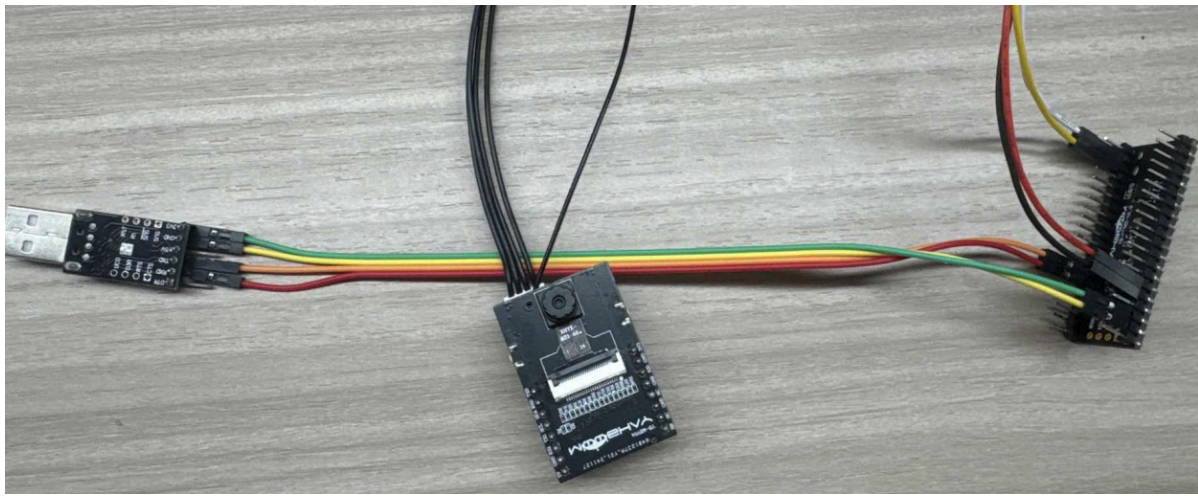
2.接线示意图

STM32	esp32摄像头
PB10	SCL
PB11	SDA
GND	GND
5V	5V

STM32	ESP32摄像头
PA9	RX
PA10	TX
GND	GND
5V	5V



实物连接图:



3.实验步骤和实验效果

1. 检验程序是否正常运行
2. 在主函数，修改成对应的ai模式。

```
//模式选择 Model Select  
typedef enum Model_state_t  
{  
    Normal=0x00,  
    Cat_Dog_Model, ←  
    Face_Detection,  
    Color_identify,  
    Face_identify,  
    QR_code,  
    MAX_ERROR  
}Model_state;
```

```
Model_state model = Cat_Dog_Model;
```

3. 把本工程的程序下载到STM32板子上
4. 在电脑打开串口助手，打开电脑检测出STM32的串口，如下图



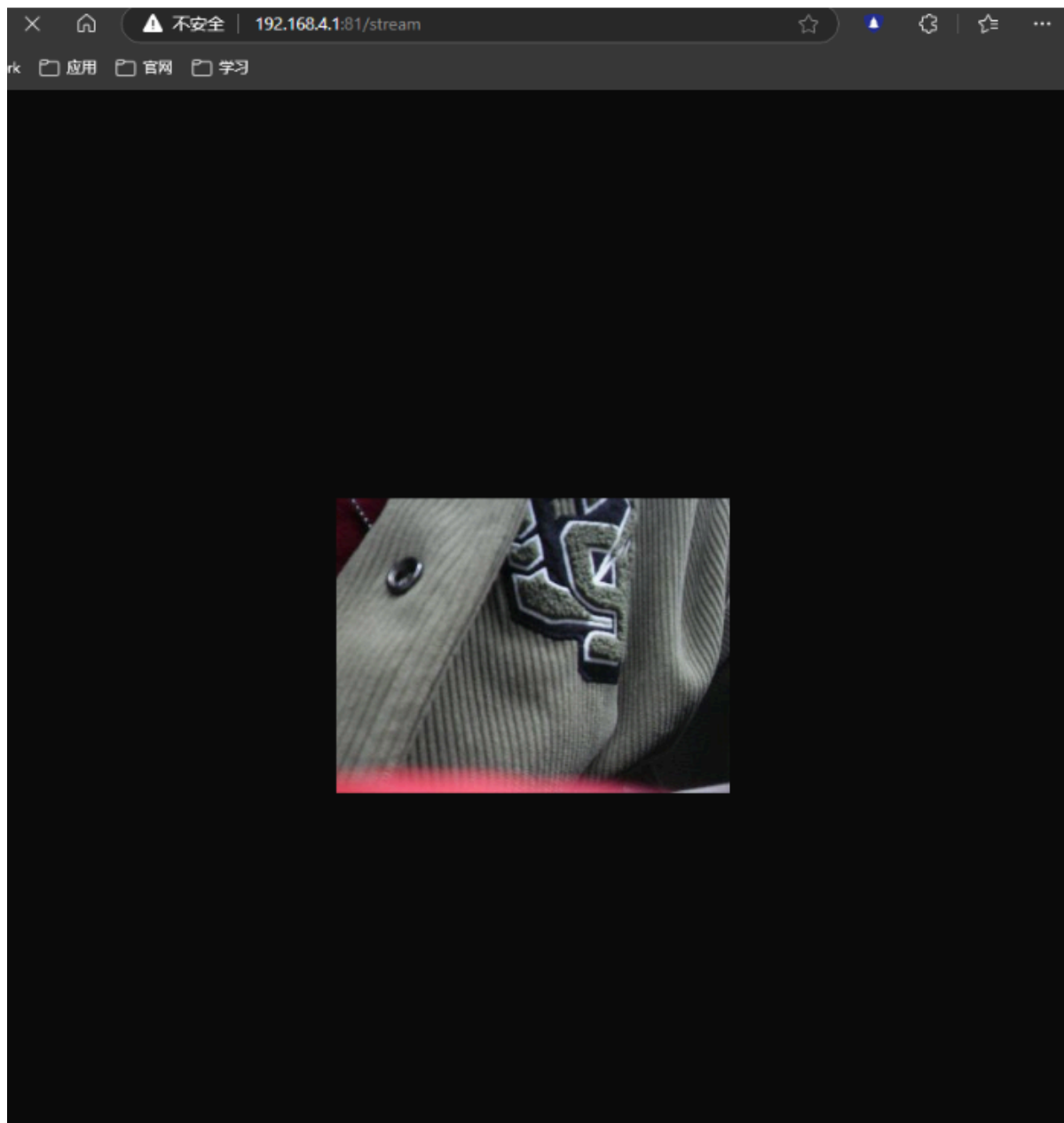
按下STM32的复位键后，串口助手将打印出对应的信息



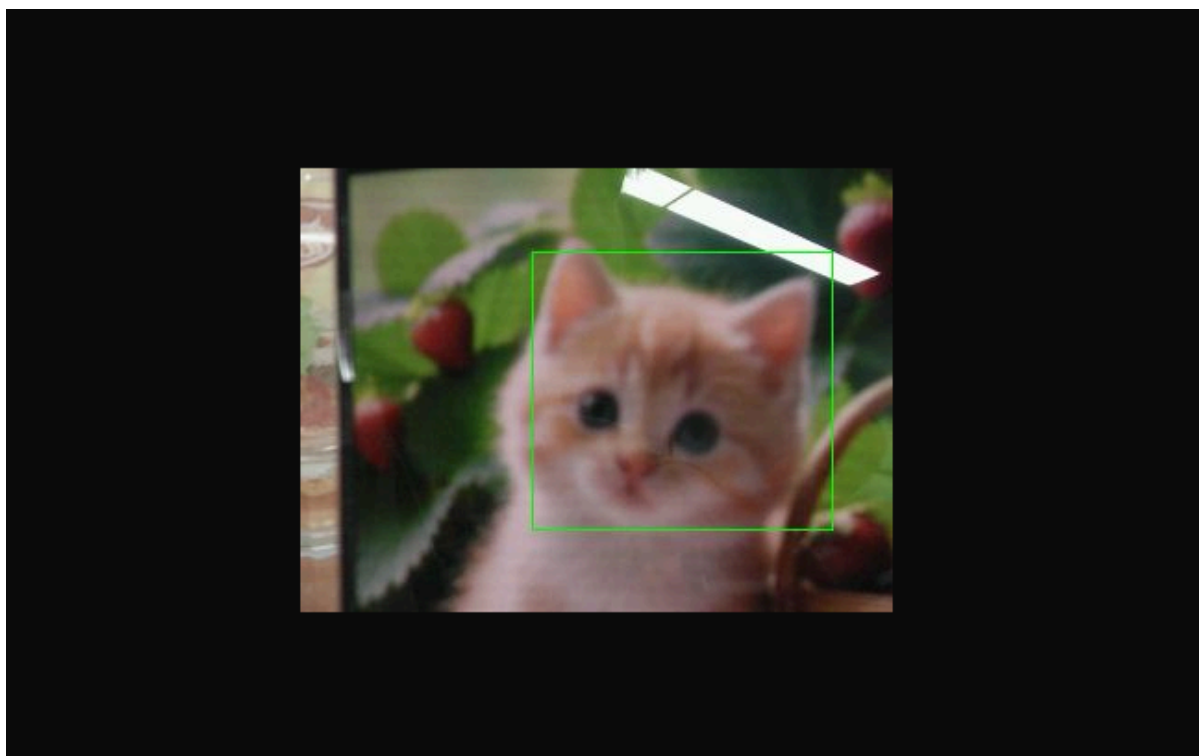
4.打开摄像头，连接上esp32发布出的热点



然后通过浏览器输入 <http://192.168.4.1:81/stream> 这个访问摄像头画面



5.识别猫狗。识别成功会打印出当前的中心坐标，把猫的图片放到刚刚的画面前。



同时终端会打印出当前的坐标,

```
[2024-12-27 19:37:46.944]# RECV ASCII>
mx:189  my:104  area:16300  id:0

[2024-12-27 19:37:47.241]# RECV ASCII>
mx:189  my:105  area:16564  id:0

[2024-12-27 19:37:47.520]# RECV ASCII>
mx:190  my:106  area:16335  id:0

[2024-12-27 19:37:47.814]# RECV ASCII>
mx:192  my:108  area:14256  id:0

[2024-12-27 19:37:48.094]# RECV ASCII>
mx:203  my:112  area:14926  id:0

[2024-12-27 19:37:48.395]# RECV ASCII>
mx:204  my:111  area:27982  id:0
```

人脸识别模式

当切换成人脸识别模式时,

```
#include "AllHeader.h"

Model_state model = Face_identify;
```

编译下载到STM32主板上, 打开串口调试助手



```
[2024-12-30 10:24:46.757]# RECV ASCII>
mx:160  my:120  area:0  id:0

[2024-12-30 10:24:47.008]# RECV ASCII>
mx:160  my:120  area:0  id:0

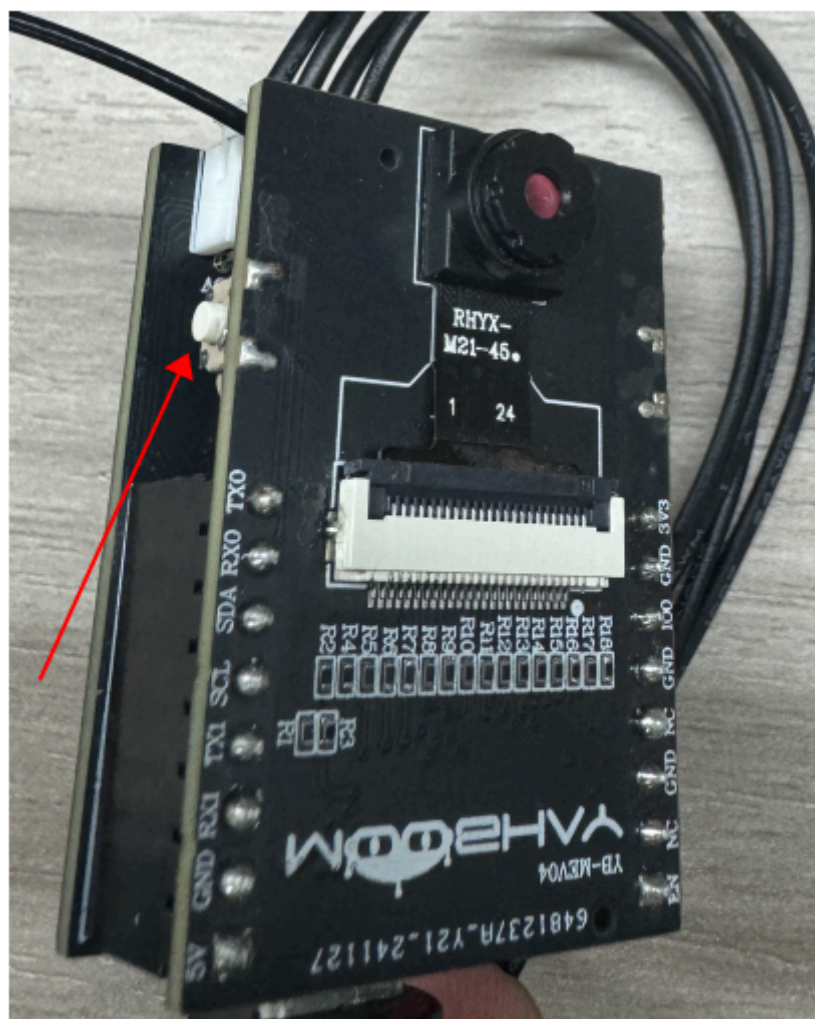
[2024-12-30 10:24:47.242]# RECV ASCII>
mx:160  my:120  area:0  id:0

[2024-12-30 10:24:47.475]# RECV ASCII>
mx:160  my:120  area:0  id:0

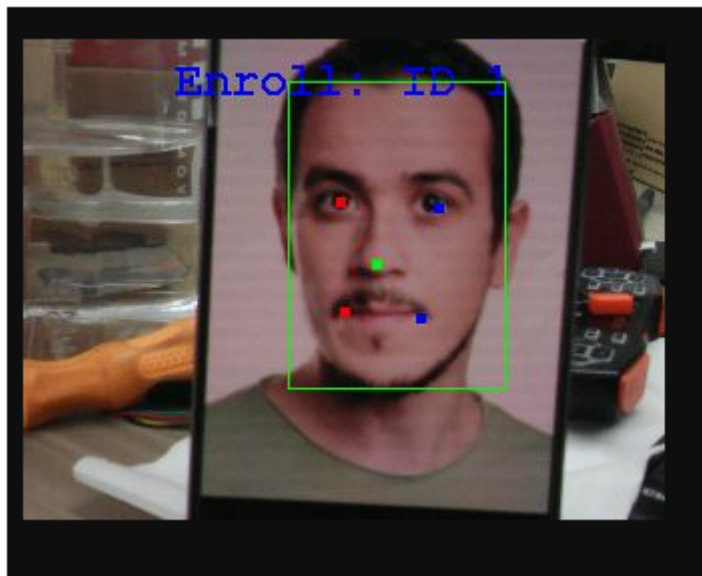
[2024-12-30 10:24:47.726]# RECV ASCII>
mx:160  my:120  area:0  id:0

[2024-12-30 10:24:47.962]# RECV ASCII>
mx:160  my:120  area:0  id:0
```

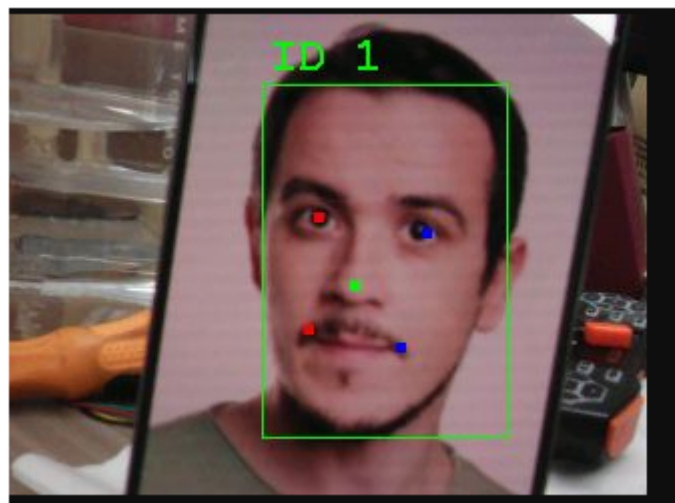
识别人脸。当时别到人脸的时候按下key按键表示记录人脸



出现下图就是记录成功了，记录成了人脸1



这时候可以长按按键两秒，之后松开再按一下按键就能识别当前的人脸



同时终端会打印出当前的中心坐标，以及识别到的人脸。

```
[2024-12-30 10:46:41.234]# RECV ASCII>
mx:134  my:135  area:25821  id:1

[2024-12-30 10:46:41.566]# RECV ASCII>
mx:134  my:135  area:25821  id:1

[2024-12-30 10:46:41.877]# RECV ASCII>
mx:134  my:135  area:25821  id:1

[2024-12-30 10:46:42.191]# RECV ASCII>
mx:134  my:135  area:25821  id:1

[2024-12-30 10:46:42.520]# RECV ASCII>
mx:134  my:135  area:25821  id:1

[2024-12-30 10:46:42.834]# RECV ASCII>
mx:134  my:135  area:25821  id:1
```

颜色检测模式

当切换到颜色检测模式时，

```
Model_state model = Color_identify;
```

编译下载到STM32主板上，打开串口调试助手



```
[2024-12-30 10:24:46.757]# RECV ASCII>
mx:160  my:120  area:0  id:0

[2024-12-30 10:24:47.008]# RECV ASCII>
mx:160  my:120  area:0  id:0

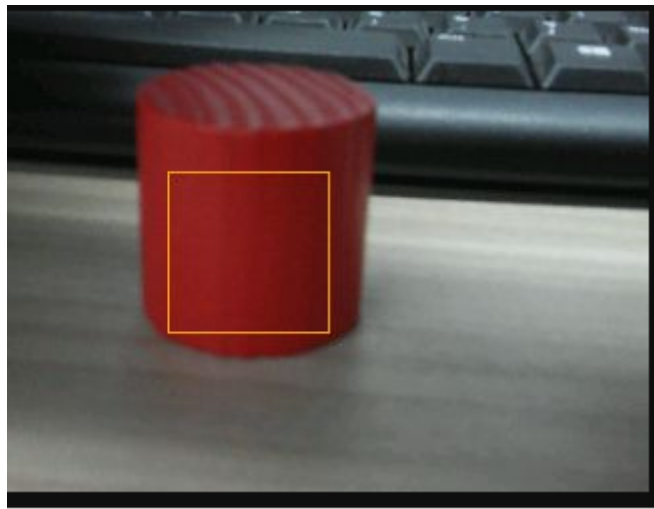
[2024-12-30 10:24:47.242]# RECV ASCII>
mx:160  my:120  area:0  id:0

[2024-12-30 10:24:47.475]# RECV ASCII>
mx:160  my:120  area:0  id:0

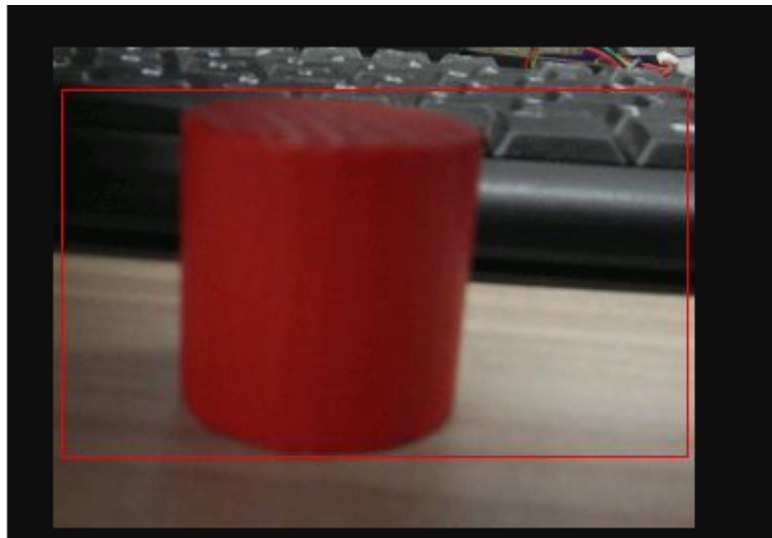
[2024-12-30 10:24:47.726]# RECV ASCII>
mx:160  my:120  area:0  id:0

[2024-12-30 10:24:47.962]# RECV ASCII>
mx:160  my:120  area:0  id:0
```

识别颜色。按一下按键，会出现一个框，可以用这个框来框选要用的颜色。



再长按按键两秒，松开再按一次就能识别当前框选的颜色，同时出现红框，



同时终端会打印出当前的中心坐标，

```
[2024-12-27 19:37:46.944]# RECV ASCII>  
mx:189  my:104  area:16300  id:0  
  
[2024-12-27 19:37:47.241]# RECV ASCII>  
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mx:190  my:106  area:16335  id:0  
  
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[2024-12-27 19:37:48.395]# RECV ASCII>  
mx:204  my:111  area:27982  id:0
```

